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From burden to beacon: Mapping Chinese students' L2 shame, guilt, motivation, and gender in Informal Digital Learning of English (IDLE)

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ABSTRACT

Purpose: While emotions and motivations influence second language (L2) learning, little is known about roles of self-conscious emotions, specifically L2 shame and guilt, in informal digital learning of English (IDLE). This study aims to examine how L2 shame and guilt interact with L2 motivation to influence learners' IDLE engagement across gender.

Design/Methodology: 2,168 questionnaires were collected from Chinese undergraduate English as a Foreign Language (EFL) students. Structural equation modeling was employed to explore the relationships between L2 shame, L2 guilt, ideal and ought-to L2 selves, and IDLE and identify gender's moderation.

Findings: Among these students, those with greater L2 shame had stronger ought-to L2 selves but weaker ideal L2 selves, both of which, in turn, yielded less IDLE engagement. By contrast, students with greater L2 guilt had both stronger ought-to L2 selves and stronger ideal L2 selves, both of which, in turn, yielded more IDLE. Relative to males, females reported more L2 shame and L2 guilt. Compared to males, females with stronger sense of L2 shame showed less IDLE, fully mediated by weaker ideal L2 self and stronger ought-to L2 self. Relative to males, females with greater L2 guilt had more IDLE, partially mediated by stronger ideal L2 self and stronger ought-to L2 self.

Originality/value: This is the first study to integrate negative self-conscious emotions and gender into the L2 motivation to shape learners' IDLE engagement. It offers valuable insights for stakeholders seeking to cultivate supportive, shame-free learning environments, harness constructive aspects of guilt, implement gender-sensitive strategies, and promote IDLE activities.

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1. Introduction

As technology spreads across the globe, more students are leveraging it to learn a second language (L2) beyond classrooms (Lai and Wang 2024; Reinders and Benson 2017). A concept that has gained momentum through the use of digital technologies in informal settings is Informal Digital Learning of English (IDLE). Lee (2019, 768) defines IDLE as ‘self-directed, self-guided, and autonomous language learning activities (e.g. browsing English news websites) by harnessing a diverse array of digital devices (e.g. mobile phones and laptops) and resources (e.g. webpages and social media apps).’ Many studies have investigated that participating in IDLE practices for language learners can not only provide a series of linguistic and psychological benefits but also facilitate the improvement of intercultural competence (Lee 2020). IDLE can improve language proficiency (Lee and Dressman 2018; Zou et al. 2025d), boost language enjoyment (Liu et al. 2025), lessen language anxiety and boredom (Zou et al. 2025b, 2025g), facilitate willingness to communicate (Lee and Lu 2023; Zou et al. 2025b, 2025g), and promote self-efficacy and grit (Barkati et al. 2024; Liu, Zhang, and Zhang 2024c). Recent research has increasingly acknowledged the pivotal role of individual differences, particularly motivational and affective factors, in shaping learners’ engagement with IDLE practices (Liu, Zhang, and Zhang 2024c; Zou et al. 2025c, 2025e).

Previous studies have comprehensively investigated the influence of motivational factors, especially the L2 motivational self-system, on learners’ participation in IDLE activities (Lee and Lu 2023; Liu, Darvin, and Ma 2024a). For example, playing digital games as a type of IDLE activity can be positively associated with ideal L2 selves and ought-to L2 (Dörnyei 2009; Soyoof et al. 2023). The ideal L2 self encompasses the learner’s aspiration of becoming a proficient user of the second language, motivated by hopes and ambitions (Dörnyei 2009). The ought-to L2 self, on the other hand, is how the learner thinks they should be in the L2, which is often based on social norms and the desire to avoid bad outcomes (Dörnyei 2009). Among affective variables, a number of studies (e.g. Teimouri 2017; 2018; Yan, Sun, and Shi 2024) reported that L2 shame and L2 guilt as self-conscious emotional factors can empower learning behaviors and academic engagement through their enhancement of learners’ sense of accomplishment. To date, however, there is a scarcity of study on the association between IDLE activities and L2 shame and L2 guilt.

The importance of L2 shame and L2 guilt in IDLE contexts is manifested across several crucial dimensions. As self-conscious emotions, L2 shame and L2 guilt distinctively emerge from personal achievements and parental and social recognition in language learning. In contrast to more general and situation-dependent affective variables such as enjoyment, boredom, and burnout, L2 shame and L2 guilt specifically link learners’ sense of self-esteem to their language learning trajectory. This unique characteristic positions L2 shame and L2 guilt as especially relevant for understanding motivated behaviors and involvement in IDLE contexts. The nexus between L2 shame, L2 guilt, and L2 motivation is firmly grounded in both the Control-Value Theory of achievement emotions (Pekrun 2006) and the L2 Motivational Self System (Dörnyei 2009). As self-conscious achievement emotions, L2 shame may lessen learners’ learning motivation, while L2 guilt may reinforce their ideal L2 selves and ought-to L2 selves and facilitate involvement in IDLE activities aligned with their language learning aspirations and social expectations. This theoretical foundation elaborates the conceptual significance of L2 shame and L2

guilt in negotiating motivation and fostering engagement within digital and technological learning contexts. Beyond their immediate relevance to IDLE contexts, this investigation of L2 shame and L2 guilt carries broader implications for L2 learning and educational psychology. It enriches the growing literature on emotions in language learning (Dewaele et al. 2016; Zou et al. 2025a, 2025f, 2025g). This positioning within wider disciplinary contexts underscores the study's significance in exploring the novel dimension of L2 shame and L2 guilt.

The gap in understanding the interplay between L2 shame, L2 guilt, and motivation assumes paramount importance in the research context of IDLE, where successful IDLE learners must sustain their participation in English practices autonomously and proactively (Lee 2019; Lee and Dressman 2018; Liu, Zou et al. 2025). On this ground, while IDLE presents unprecedented opportunities for self-directed learning, it also necessitates heightened emotional factors and sustained motivation from learners. Elucidating the contributions of L2 shame and L2 guilt to this emotional foundation and motivational dynamics could yield pivotal insights for developing more effective pedagogical interventions that equip learners for autonomous participation in IDLE.

Moreover, gender differences are widely recognized as a crucial individual difference factor in EFL learning and have greatly gained researchers' attention over recent decades (e.g. Yao, Zhu, and Zhan 2024). Recent literature has revealed notable gender differences in IDLE and emotional experiences (e.g. Else-Quest et al. 2012; Kushnir et al. 2016; Lee, Xie, and Lee 2021; Liu, Zou et al. 2025). For instance, gender has been shown to affect Korean learners' IDLE behaviors in terms of alleviating L2 anxiety (e.g. Lee, Xie, and Lee 2021). Gender also was found to influence learners' general shame and guilt (e.g. Lutwak and Ferrari 1996). By surveying 207 participants, Kushnir et al. (2016) found that females exhibited stronger shame and guilt proneness than males. Else-Quest et al. (2012) conducted a meta-analysis study related to self-conscious moral emotions (including shame, guilt, embarrassment, and pride) and found that general shame and guilt proneness demonstrated small gender differences, whereas embarrassment and pride displayed gender similarities. However, little is known about how gender influences the relationship between self-conscious emotions (i.e. L2 shame and L2 guilt) and IDLE behaviors in L2 settings. Thus, there is a pressing need to better understand the role of gender in the links between L2 shame, L2 guilt, and IDLE engagement. To address these research gaps, this study examines the following three questions:

RQ1: To what extent, if any, do L2 shame and L2 guilt predict L2 motivation?

RQ2: To what extent, if any, do L2 shame and L2 guilt interact with L2 motivation to predict IDLE engagement?

RQ3: Does gender play a moderating role in the relationships between these variables?

2. Literature review

2.1. Informal digital learning of English (IDLE)

Rooted in the crossroads of computer-assisted language learning (CALL) and language learning out-of-class (Dressman and Sadler 2020; Lai and Wang 2024; Lee 2019; Soyoof et al. 2023; Sundqvist and Uztosun 2024), IDLE has attracted L2 researchers' great

attention. IDLE is a hierarchical factor with form-focused and meaning-focused IDLE parts (Zou et al. 2025c, 2025d) that activate different cognitive pathways in students' brains (Lai, Zhu, and Gong 2015). Form-focused IDLE activities require students to think about linguistic structures and forms like grammar (e.g. grammar exercises, vocabulary drills; Zadorozhnyy and Lee 2023), thus sharpening their accuracy in language use (Liu, Darwin, and Ma 2024a; Zou et al. 2025c, 2025d). By contrast, meaning-focused IDLE practices focus on understanding and expressing meaning (e.g. listening to a conversation, reading the transcript of an interview with a scientist; Lai and Wang 2024). Such activities boost students' comprehension and fluency (Zadorozhnyy and Lee 2023). Balanced integration of form-focused and meaning-focused activities improved students' L2 learning outcomes and confidence (Zou et al. 2025c, 2025d).

2.2. Theoretical framework

Dornyei's (2009) L2 Motivational Self System (L2MSS) is recognized as one of the most predominant and influential theories for understanding L2 motivation dispositions (Li, Hiver, and Papi 2022; You, Dornyei, and Csizer 2016). Its two sub-constructs – the ideal L2 self and the ought-to L2 self – have gained great attention in diverse EFL contexts (Lee and Lu 2023; Liu, Darwin, and Ma 2024b; Liu, Zou et al., 2025). Dornyei (2009) points out that the ideal L2 self can drive learners 'to learn the L2 because of the desire to reduce the discrepancy between [their] actual and desired future L2 selves' (29), while the ought-to L2 self refers to motivate learners to study L2 to meet their peripheral expectations (e.g. parents, teachers, and peers) and avoid adverse consequences (e.g. examination failure). Also, the ideal L2 self contains integrative motives (e.g. becoming a member of the L2 community) and internalized instrumental motives (e.g. pursuing a satisfactory career) in the social-educational model (Gardner 2010). In this construct, the ideal L2 self is promotion-oriented, concentrated on achieving aspirations, and aiming at success. In contrast, the ought-to L2 self is prevention-oriented, anchored on meeting external expectations and avoiding negative outcomes (Dornyei 2009; Sugita McEown, Noels, and Chaffee 2014). Such learners are motivated extrinsically, and the instrumentality is mostly external rather than internalized (Sugita McEown, Noels, and Chaffee 2014; Yung and Chiu 2023).

More importantly, it is worth noting that L2MSS is facing ongoing scholarly debate regarding its conceptual robustness and measurement validity, especially about the discriminant validity of scales (e.g. Al-Hoorie and Hiver 2025; Papi and Teimouri 2024). Drawing on correlational and factor-analytic data from 384 South Korean secondary school students, Al-Hoorie and Hiver (2024) argued that the framework suffers from issues of discriminant validity and suggested reconsidering its continued application in L2 motivation research. In response to their assertion, Papi and Teimouri (2025) refuted their assertion from factor analytic results (i.e. exploratory factor analysis and confirmatory factor analysis) and methodological issues (including issues in the design and implementation of Al-Hoorie and Hiver's (2024) study, inappropriately run data analysis, and an issue with a 7-point Likert scale with a neutral midpoint concerns the conceptual ambiguity of the midpoint) by examining the evidence demonstrated and re-analyzing Al-Hoorie and Hiver's (2024) data. In addition, Papi and Teimouri (2024) emphasized the importance of 'establishing the definitional validity of different constructs and

operationalizing them based on agreed-upon definitions and the context of each study' (10). In response to this call, the present study carefully considered the conceptualizations of the ideal L2 self and ought-to L2 self, providing clear and contextually grounded definitions to ensure definitional validity.

While L2MSS primarily addresses self-images and future-oriented selves, the Control-Value Theory (CVT) of achievement emotions (Pekrun 2006) offers a complementary perspective by highlighting the role of affective factors in shaping motivation and learning behaviors. The CVT shows how emotions' three dimensions of valence (unpleasant or pleasant), activation (deactivate or activate), and object focus (activity-oriented or outcome-oriented) shape motivation and learning. Valence refers to the perceived pleasantness or unpleasantness of an emotional experience; activation denotes the extent of emotional arousal, encompassing both physiological and psychological responses; and object focus relates to the locus of the emotional response – whether it is directed toward the process of engaging in a task (i.e. activity-focused) or the results achieved from the task (i.e. outcome-focused) (Pekrun 2006). Following this taxonomy, L2 shame is described as unpleasant, activating, and outcome-oriented, primarily relating to the retrospective outcomes of achievement-related activities or tasks. This suggests that L2 shame often arises from the anticipation of retrospective failure outcomes (Leary 2007; Lewis 1971). However, while L2 guilt is also unpleasant and activating, it may not solely be associated with the retrospective outcomes of an activity (outcome-oriented). L2 guilt can also be related to the process of an ongoing achievement-related activity (activity-oriented). L2 guilt often stems from an individual's perception that they have not fulfilled their responsibilities in their past or present actions, thus potentially influencing prospective outcomes (Pekrun and Perry 2014; Tangney and Dearing 2003).

Shame and guilt are two unpleasant, activating, retrospective outcome emotions (Pekrun 2006; Pekrun and Perry 2014). Both are also self-conscious, social, and outcome-oriented emotions (Tracy, Robins, and Tangney 2007). As students who feel L2 shame (e.g. from a low-test score) fear failing and being seen as a failure, they often anticipate harsh judgments from their expected audience, so they shrink back, hide, and avoid trying, speaking up, or joining in (Leary 2007; Lewis 1971; Tracy, Robins, and Tangney 2007). By contrast, L2 guilty students scrutinize their mistakes, often feeling tension, regret, and remorse (Tangney and Dearing 2003), accept responsibility for them, and try to fix them in L2 settings (Teimouri 2018, 2025). Students feeling L2 guilty about a low grade often ascribe the cause to their insufficient effort rather than a lack of ability and strive to compensate for their shortcomings in L2 settings (Teimouri 2018). Hence, L2 shame is more painful and devastating than L2 guilt (Teimouri 2018). While L2 shame freezes students' learning behavior, L2 guilt drives them forward or engage with learning activities (Lewis 1971; Teimouri 2025). IDLE to some extent can be regarded as learners' engagement behaviors, which features diverse language learning activities with the affordance of digital technologies (Liu and Ma 2024; Zadorozhnyy and Lee 2023).

Drawing on Dornyei's (2009) L2 Motivational Self System (L2MSS) theory and Pekrun's Control-Value Theory (CVT) (2006), during IDLE activities such as watching L2 movies and reading L2 news, students might face two scenarios. In the first one, students are watching an L2 movie but do not understand the dialogue. Focusing on the outcome, they see themselves as a failure and feel a strong sense of L2 shame (Pekrun 2006). Feeling this

painful self-threat (negative valence), they focus on the failure outcome, which reduces their sense of their potential (deactivation; Pekrun 2006). Thus, they might downgrade their ideal and ought-to L2 selves (weaker L2 motivation), reduce their effort, and stop watching the L2 movies (less IDLE engagement; Soyoof et al., 2023; Zou et al. 2025c, 2025d). In the second scenario, students are reading L2 news on a website but do not understand it. They feel the pain of inadequate preparation and self-blame (negative valence) but their focus on the changeable process yields high arousal (activation, Pekrun 2006) and can boost their ideal and ought-to L2 selves (stronger L2 motivation). As such, they redouble their efforts and look up unfamiliar words from the news in an online dictionary out-of-class (more IDLE engagement; Soyoof et al., 2023; Zou et al. 2025c, 2025d). Thus, the L2MSS and Control-Value Theory of achievement emotions provide a solid foundation for elaborating the interplay between L2 shame, L2 guilt, motivation, and IDLE, facilitating the illumination of formulating the hypotheses.

2.3. Formulating the hypotheses

2.3.1. Empirical studies of L2 shame, L2 guilt, and L2 motivation

While studies exploring the interplay between L2 shame, L2 guilt, and L2 motivation are limited, several research studies have disclosed that L2 shame can exert a negative effect on learners' motivation (Shao et al. 2023; Yan, Sun, and Shi 2024), but L2 guilt had a positive influence on L2 learners' motivation (Teimouri 2018, 2019). Specifically, the relationship between L2 shame and L2 motivation has been substantiated through some empirical studies. In a study of 174 Iranian university English-major students, Teimouri (2018) investigated how L2 shame and L2 guilt predicted L2 motivation (including intended effort, L2 willingness to communicate [WTC], and attention) and language achievement. His findings demonstrated that L2 shame negatively and significantly predicted L2 motivation, which was consistent with previous discussions on the negative influence of shame on learners' motivation (Lewis 1971; Tangney and Dearing 2003; Tracy, Robins, and Tangney 2007). L2 guilt exerted a positive and significant impact on L2 motivation, which can be found that learners experiencing guilt, characterized by feelings of tension, remorse, and regret, may proactively pursue new methods to rectify or withdraw their prior misbehaviors (e.g. insufficient effort). Similarly, focusing on 402 northern Chinese university EFL learners, Yan, Sun, and Shi (2024) employed a structural equation modeling approach to examine the associations between teacher autonomy support, two under-emphasized achievement emotions (i.e. hope and shame), and intended effort (a descriptor of L2 motivation). Their research results revealed that hope positively predicted the intended effort, while L2 shame made a negative impact on the intended effort and played a mediating role in the relationship between teacher autonomy support and intended effort. Given the established relationships between L2 shame, L2 guilt, and L2 motivation in EFL contexts, we propose the following hypotheses regarding the relationship between L2 shame, L2 guilt, and L2 motivation:

- H1. Among students, those with greater L2 shame have lower ideal L2 selves.
- H2. Among students, those with greater L2 shame have lower ought-to L2 selves.
- H3. Among students, those with greater L2 guilt have greater ideal L2 selves.
- H4. Among students, those with greater L2 guilt have greater ought-to L2 selves.

2.3.2. Empirical studies of L2 shame, L2 guilt, and IDLE engagement

Research exploring the relationship between L2 shame, L2 guilt, and IDLE is scarce. Very few studies have examined L2 shame's negative influence on learners' behaviors and L2 guilt's positive impact on their behaviors (Shao et al. 2023; Teimouri 2018). Specifically, Teimouri (2018) surveyed 174 Iranian tertiary students to probe the relationships between L2 shame, L2 guilt, and learning behaviors. The study's results have displayed that L2 shame had a strong and negative influence on L2 learners' communicative and interpersonal behaviors, whereas L2 guilt had a significant and positive impact on their communicative behaviors. A comprehensive examination of the extant literature in educational psychology reveals that shame entails a negative evaluation of one's global self on the appearance of an unfavorable event. Thus, learners experiencing shame may eschew social activities or conceal themselves from others and exhibit a diminished propensity to voluntarily interact with others, thereby hampering the development of language attainment (Lewis 1971; Tracy, Robins, and Tangney 2007). By contrast, L2 guilty students direct their attention to their own behaviors and are more likely to accept responsibility for their actions (Teimouri 2018). L2 guilt pushes students to redeem themselves or prevent similar shortcomings in the future, often motivating them to pursue new methods to do so (Teimouri 2018, 2019). Such a positive attitude, exertion of effort, and expectation of future improvement can (a) boost students' perception of their potential and raise their ideal L2 self and (b) validate external expectations and strengthen their ought-to L2 selves. Hence, L2 guilt can drive motivation to engage in learning activities (Teimouri 2017), especially IDLE engagement. Hence, based on these findings of previous studies, we propose these hypotheses:

H5. Among students, those with greater L2 shame engage in fewer IDLE activities.

H6. Among students, those with greater L2 guilt engage in more IDLE activities.

2.3.3. Empirical studies of L2 motivation and IDLE engagement

The contributing role of L2 motivation and involvement in IDLE activities has been evidenced by a body of research (Liu, Darwin, and Ma 2024a, 2024b; Zou et al. 2025c), which provides empirical foundations for formulating hypotheses concerning the motivational mechanism underlying IDLE participation. Ideal L2 self and ought-to L2 self can both boost learning activities, like IDLE. Among students, those with stronger ideal L2 selves see its larger vivid differences from their actual selves (Dörnyei and Ushioda 2021). So, they aim to get closer to their ideal L2 selves by developing action plans and implementing learning strategies (Dörnyei and Ushioda 2021). As students have autonomy to readily implement accessible IDLE, those with stronger ideal L2 selves engage in more IDLE activities (Liu, Zhang, and Zhang 2024c).

H7. Among students, those with greater ideal L2 selves engage in more IDLE activities.

Students with greater ought-to L2 selves than others feel the weight of family, teacher, or societal expectations, which can fuel their dutiful L2 learning (Liu, Zhang, and Zhang 2024c; Zou et al. 2025f). When family members, teachers, or others have high expectations of students, such expectations can ignite their ought-to L2 selves, thereby directly helping students invest in IDLE activities (Liu, Darwin, and Ma 2024a; Zou et al. 2025c). For example, focusing on Chinese tertiary EFL undergraduates, Zou et al. (2025c) surveyed

1,388 students and employed structural equation modeling to investigate the interplay between L2 pride, future selves (i.e. ideal L2 self and ought-to L2 self), and IDLE engagement. Their findings displayed that Chinese students' ought-to L2 selves significantly and positively predict their IDLE engagement. Hence, peripheral expectations can also be a supportive and pivotal force, creating an environment that encourages learning (Teimouri 2017, 2025; Zou et al. 2025c, 2025f). As such, students with greater ought-to L2 selves than others engage in more IDLE activities (Reinders, Lai, and Sundqvist 2022).

H8. Among students, those with greater ought-to L2 selves engage in more IDLE activities.

2.4. Gender effects across variables

Several studies have revealed that gender differences influence L2 learners' emotions and their IDLE engagement (e.g. Lee et al. 2021; Liu, Zou et al. 2025). This insight echoes Dewaele et al. (2016), who found that girls had significantly more enjoyment in foreign language (FL) classrooms and were prouder than boys of their FL proficiency, while they were less confident in using FL than their boy peers. Concerning the self-conscious emotions, Lutwak and Ferrari's (1996) found that girls have a stronger sense of general shame proneness and guilt proneness than boys in the learning process. Compared with men, women are more sensitive to others' expectations (e.g. teachers or peers), so when they fail to meet societal expectations, they feel more shame than men do (Else-Quest et al. 2012; Kushnir et al. 2016). Compared with men, women are more sensitive to societal expectations, so those with more shame view themselves as less capable and have worse self-perceptions (Ferguson and Crowley 1997), so we expect them to have lower ideal L2 selves. In addition, compared with men, women with stronger L2 shame are more likely to reduce their ought-to L2 selves due to their sensitivity to societal and peripheral expectations (Kushnir et al. 2016). Hence, we propose the following hypotheses:

H9a. Females feel more L2 shame than males do.

H9b. The negative link between L2 shame and ideal L2 self is stronger among females than males.

H9c. The negative link between L2 shame and the ought-to L2 self is stronger among females than males.

Furthermore, female students are more prone than male students to guilt while learning (Lutwak and Ferrari 1996). Sensitive to others' expectations, women also internalize them, owning these expectations (Eisenclas 2013). Hence, compared with men, women who fail to meet these expectations often feel more guilt (Else-Quest et al. 2012; Kushnir et al. 2016). As with shame, guilt is more common among female students than male students (Lutwak and Ferrari 1996). Similarly, Kushnir et al.'s (2016) study has uncovered that females demonstrated higher level of general shame proneness and guilt proneness than male counterparts. While these studies have corroborated in general educational psychology field, more research is required to evidence trait-like shame proneness and trait-like guilt proneness in L2 settings. Following traditional societal views of masculinity that emphasize achievement, men often value their competence and achievements more, compared with women (Ridgeway 2011). Hence, when men fail to meet their achievement expectations and feel guilty, they do not lower their expectations but hold on to

them more tightly (Eisenchlas 2013). Hence, compared with women, we expect men feeling L2 guilt to have stronger ideal L2 selves and ought-to L2 selves.

H10a. Females feel more L2 guilt than males do.

H10b. The positive link between L2 guilt and ideal L2 self is stronger among males than females.

H10c. The positive link between L2 guilt and ought-to L2 self is stronger among males than females.

Gender differences also extend to IDLE engagement in the L2 context (Lee, Xie, and Lee 2021; Liu et al., 2025; Sundqvist and Sylvén 2014). For instance, Sundqvist and Sylvén (2014) explored whether gender differences influence L2 learners to engage with different digital extramural activities. Their research findings demonstrated that females are more likely to invest time in facebooking and spend significantly more time on pastime language-related activities in Swedish than male peers. However, boys have a greater preference for playing digital games and watching films than girls. In a similar vein, a case by Lee, Xie, and Lee (2021) has demonstrated that females were influenced by IDLE engagement associated with alleviating L2 anxiety compared with males. By contrast, Liu and his colleagues (2025) found that gender differences did not influence the link between Artificial Intelligence (AI)-mediated IDLE and foreign language enjoyment. The inconsistent results concerning IDLE engagement indicated that more research needs to be conducted to better understand the role of gender in IDLE engagement. Moreover, there was no study that investigated a moderating role of gender in the direct and mediated relationships between L2 shame, L2 guilt, and IDLE engagement. Hence, we test the following hypothesis.

H11. IDLE engagement does not differ across gender.

Taken together, the hypothesized structural model is in Figure 1.

3. Methodology

To examine these hypotheses, a survey was administered to undergraduate students in China. We used structural equation modeling (SEM) to analyze the questionnaire data.

3.1. Research context and participants

Situated in the Chinese university EFL context, this study recruited participants from 15 public universities in South and Central China through online channels. Employing a convenience sampling technique (Creswell and Crewell 2017), our team circulated an e-poster of an online questionnaire incorporating a QR code via the WeChat social media platform. Collaborating with 45 university teachers who helped to disseminate the survey to their students between May and December 2024. These teachers were introduced to understand the research aims and subsequently explained the purpose of the study to their students.

A total of 2,168 students participated in the survey. All participants met the following inclusion criteria: (1) being currently enrolled as undergraduate students at a Chinese

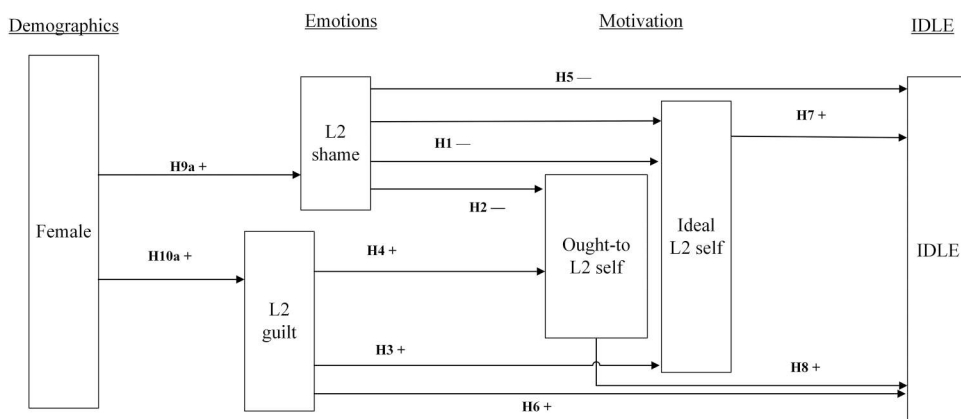


Figure 1. The hypothesized structural model. Note: The ‘Female’ parameter is relative to ‘Male.’ Due to the complex moderating roles of gender across the relationships among variables, the corresponding hypotheses were not presented in this diagram.

university and (2) self-identifying as English as a Foreign Language (EFL) learners. Participants gave their consent for anonymity and voluntary participation. Anonymity was used to protect their identities. The study also obtained the ethics approval of the last authors’ institution, adhering to the ethical guidelines of China.

3.2. Survey measures

The survey instrument was translated between English and Chinese using a forward and back-translation procedure to ensure accuracy (see Appendix A). It comprised demographic items (e.g. gender, age, academic major, ownership of digital devices, and Wi-Fi availability) and five psychometric scales. Participants responded to each item using a six-point Likert-type scale, ranging from 1 (strongly disagree) to 6 (strongly agree).

- a. L2 shame and L2 guilt scales (26 items): To measure the trait-like L2 shame and L2 guilt proneness of Chinese EFL students, the present study utilized the L2 Shame and L2 Guilt scales previously developed and validated by Teimouri (2018). Teimouri’s (2018) Second Language Test of Shame and Guilt Affect (L2-TOSGA) scale posed potential L2 shame scenarios (e.g. ‘In class, the teacher asks you a question in English. You don’t understand the question. The teacher asks the question in a different way. Again, you don’t understand the question.’). Then, respondents rated statements capturing *trait-like L2 shame proneness* (13 items, Cronbach’s $\alpha = .96$, e.g. ‘I would definitely feel ashamed’). In addition, Teimouri’s (2018) L2-TOSGA scale also posed potential L2 guilt scenarios (e.g. ‘You receive your English class report card. You are surprised to see that your grades are lower than what you expected. I should have studied more; I would try to do better in the future’). Then, respondents rated statements capturing *trait-like L2 guilt proneness* (13 items, $\alpha = .97$, e.g. ‘I would feel bad but would keep trying’).
- b. L2 motivation scale (12 items): The L2MSS scales, originally developed by Dornyei (2009) and subsequently validated by You, Dornyei, and Csizer (2016) for Chinese

students, were used to measure the ideal and ought-to L2 selves of Chinese EFL students. This scale also was validated by Liu, Zhang, and Zhang (2024c) in the informal digital language learning contexts. It included *Ideal L2 Self* (6 items, $\alpha = .94$, e.g. 'I can imagine myself talking in English with foreign friends in informal situations in the future.') and *Ought-to L2 Self* (6 items, $\alpha = .89$, e.g. 'I consider learning English important because I want to be recognized by my teachers and classmates').

- c. IDLE scale (14 items): The IDLE scale was developed and validated by Zou et al. (2025d), which measured its *form-focused IDLE* (7 items, $\alpha = 0.94$, 'I actively use mobile phone apps or websites to learn English grammar outside of class'). It also captured *meaning-focused IDLE* (7 items, $\alpha = 0.92$, 'I actively read English articles online or English-related posts on social media').

3.3. Data analysis

After screening the data for outliers, missing values, and invalid responses (i.e. completion time < 1 min), we assessed data normality (item skewness and kurtosis) and computed descriptive statistics (M, SD) and reliability (Cronbach's α) using SPSS 29.0.

A measurement model consisting of five factors and their respective observed variables was used to strengthen the construct validity of the modified questionnaire in AMOSS. Eight indices were employed to determine the degree of model fit between the measurement model and the dataset: the ratio of χ^2 to Degrees of Freedom ($\chi^2/df < 5$), Comparative Fit Index (CFI > .90), Incremental Fit Index (IFI > .90), Tucker-Lewis index (TLI > .90), Normed Fit Index (NFI > .90), Parsimony Goodness-of-Fit Index (PGFI > .50), Root Mean Square Error of Approximation (RMSEA < .10), and Standardized Root Mean Squared Residual (SRMR < .08) (Kline 2023).

Then, we tested the SEM of these explanatory variables: gender, L2 shame, L2 guilt, ideal L2 self, ought-to L2 self, and IDLE. We investigated the mediating roles of the ideal L2 self and ought-to L2 self in the relationships between L2 shame, L2 guilt, and IDLE using bootstrapping procedures with 2,000 resamples and 95% confidence intervals. To examine the moderating role of gender in these associations, we conducted a two-group comparison analysis. The *total effect* (TE) of an independent variable on the dependent variable was calculated as the sum of its direct and indirect effects. The indirect effect of explanatory variable X on outcome Y via mediator M (namely, $X \rightarrow M \rightarrow Y$) was the product of the standardized parameter linking X to M multiplied by the total effect of M on Y (namely, $[X \rightarrow M] * [M \rightarrow Y]$).

4. Research findings

After reporting descriptive statistics, we examine the data's reliability and validity. Then, we report the structural and mediation model results and moderation results.

4.1. Descriptive statistics

Altogether, 2,168 undergraduates (34% male, 66% female) completed the survey (see summary statistics in Table 1). Of these, 14% of respondents were less than 18 years old, 83% of them were between 18 and 20, and 3% were 21–22. Their majors were

Table 1. Demographics.

Variable	Category	%
Gender	Male	34
	Female	66
Age	< 18	14
	18–20	83
	21–22	3
	Other	14
Study program	Arts and humanities	28
	Social sciences	12
	Science, technology, engineering, and mathematics (STEM)	33
	Business and management	10
Ownership of types of digital devices (e.g. smartphone, laptop, etc.)	Health sciences	3
	Other	14
	1	15
	2	70
	3	12
Stable Wi-fi	More than 3	3
	Yes	72
Sufficient mobile data monthly	Yes	91

33% science, technology, engineering, and mathematics (STEM); 28% arts and humanities (e.g. English); 12% social sciences; 10% business and management; 3% health sciences; and 14% other. Regarding ownership of types of digital devices (smartphone, laptop, etc.), 70% of participants had 2 devices, 14% had 1 device, 12% had 3 devices, and 3% had more than 3 devices. Most participants (72%) had stable Wi-Fi, but 28% did not. Also, 91% respondents had sufficient mobile data, but 9% did not. Introducing each set of statistics is to establish the representativeness of our sample and enable readers to assess the transferability of findings to similar populations, thereby strengthening the generalizability of the findings across different academic disciplines.

The means of L2 shame items ranged from 3.27 to 3.71, while L2 guilt items spanned 4.40–4.51. The means of ideal L2 self items varied from 3.41 to 4.20, while the ought-to L2 items ranged from 3.44 to 3.84. IDLE is a hierarchical factor linked to its form-focused IDLE factor (.94) and its meaning-focused IDLE factor (.92). The mean of form-focused IDLE items spanned 3.59–4.24, while the meaning-focused IDLE items varied from 3.17 to 4.11. Their standard deviations showed acceptable variability (0.96–1.30), as did their kurtosis ($< |8|$) and skewness ($< |3|$; Kline 2023), indicating acceptable normality.

4.2. Reliability and validity checking

As shown in Table 2, the Cronbach's α values of eight variables were .96 (Shame), .97 (Guilt), .94 (Ideal L2 Self), .89 (Ought-to L2 Self), .94 (Form-focused IDLE), and .92 (Meaning-focused IDLE), which all outperformed the recommended threshold of .7 (Kline 2023), thus indicating that the internal consistency was achieved.

Convergent and discriminant validity were assessed to check the validity of the factors. According to Collier (2020), composite reliability (CR) and average variance extracted (AVE) were used to measure convergent validity. Table 3 showed that the CRs of the six factors exceeded the benchmark value of .7, and the AVEs were higher than the cut-off value of .5 (Hair et al. 2009). Discriminant validity was examined by comparing

the square root of AVE for each factor with their respective inter-factor correlation coefficients (Hair et al. 2009). Table 3 demonstrated that the square roots of AVE were all higher than their corresponding correlation coefficients, indicating that discriminant validity was established. Finally, our measurement and structural models fit the data well (see Table 4 and Figure 2).

Table 2. Descriptive statistics.

Factors	Items	Mean	SD	Kurtosis	Skewness	Factor loading (In CFA)	α (> .7)
L2 Shame	S1	3.59	1.23	−0.48	−0.20	.76	.96
	S2	3.32	1.23	−0.57	0.05	.80	
	S3	3.59	1.23	−0.49	−0.18	.77	
	S4	3.54	1.24	−0.58	−0.20	.77	
	S5	3.57	1.25	−0.55	−0.17	.80	
	S6	3.35	1.26	−0.64	−0.01	.72	
	S7	3.56	1.26	−0.61	−0.18	.82	
	S8	3.38	1.29	−0.67	0.06	.86	
	S9	3.49	1.25	−0.65	−0.11	.88	
	S10	3.71	1.30	−0.56	−0.28	.85	
	S11	3.27	1.24	−0.62	0.10	.87	
	S12	3.30	1.25	−0.66	0.08	.81	
	S13	3.61	1.28	−0.61	−0.19	.71	
L2 Guilt	G1	4.46	1.00	1.43	−0.91	.83	.97
	G2	4.46	0.99	1.80	−1.01	.84	
	G3	4.45	0.98	1.52	−0.91	.83	
	G4	4.51	0.97	1.98	−1.04	.82	
	G5	4.50	0.96	1.73	−0.96	.84	
	G6	4.40	1.01	1.33	−0.90	.85	
	G7	4.51	0.97	1.83	−0.99	.88	
	G8	4.46	0.99	1.62	−0.99	.88	
	G9	4.49	0.98	1.87	−1.02	.90	
	G10	4.40	1.01	1.33	−0.90	.86	
	G11	4.40	1.01	1.49	−0.93	.89	
	G12	4.46	0.98	1.23	−0.83	.91	
	G13	4.45	0.99	1.40	−0.90	.91	
Ideal L2 Self	IS1	3.56	1.27	−0.69	−0.31	.75	.94
	IS2	3.41	1.28	−0.77	−0.19	.84	
	IS3	3.61	1.27	−0.61	−0.35	.89	
	IS4	4.20	1.22	0.22	−0.77	.81	
	IS5	3.78	1.26	−0.42	−0.47	.91	
	IS6	3.91	1.26	−0.19	−0.58	.87	
Ought-to L2 Self	OS1	3.44	1.30	−0.77	−0.10	.58	.89
	OS2	3.84	1.25	−0.42	−0.47	.69	
	OS3	3.80	1.20	−0.21	−0.50	.84	
	OS4	3.68	1.21	−0.34	−0.37	.84	
	OS5	3.84	1.17	−0.01	−0.58	.83	
	OS6	3.54	1.23	−0.52	−0.26	.81	
Form-focused IDLE	F1	3.64	1.22	−0.49	−0.36	.83	.94
	F2	3.81	1.19	−0.20	−0.53	.86	
	F3	3.82	1.16	−0.09	−0.56	.87	
	F4	3.75	1.16	−0.18	−0.50	.89	
	F5	3.59	1.19	−0.48	−0.31	.86	
	F6	4.24	1.15	0.60	−0.82	.68	
	F7	3.80	1.16	−0.09	−0.51	.82	
Meaning-focused IDLE	M1	3.86	1.17	−0.07	−0.57	.75	.92
	M2	3.43	1.21	−0.62	−0.09	.87	
	M3	3.17	1.25	−0.77	0.12	.84	
	M4	3.40	1.23	−0.68	−0.14	.89	
	M5	3.21	1.24	−0.70	0.10	.86	
	M6	4.11	1.19	0.21	−0.67	.61	
	M7	3.55	1.21	−0.45	−0.22	.78	

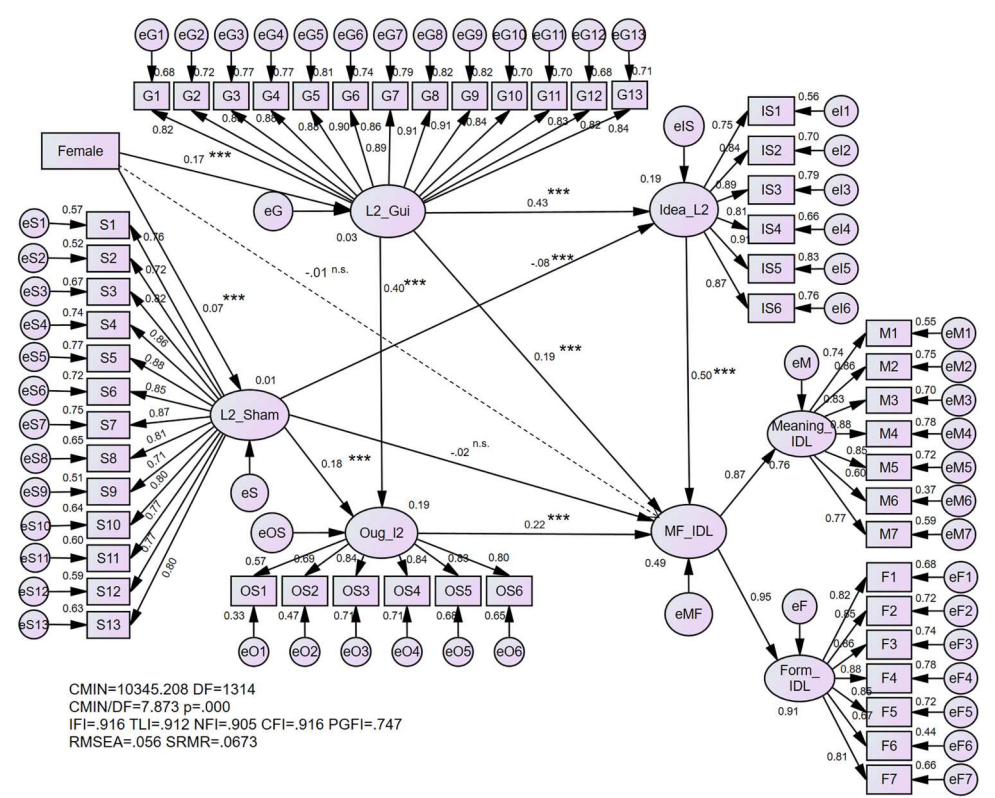


Figure 2. The structural equation model of L2 shame, L2 guilt, L2 motivation, and IDLE.
Note: The ‘Female’ parameter is relative to ‘Male.’ $p^{***} < .001$; n.s. = non significance; R2: Ideal L2 Self = 19%; Ought-to L2 Self = 19%; IDLE = 49%.

4.3. The structural equation model and mediation model measurement

The SEM supported seven hypotheses (see Table 5). Students with more L2 shame showed weaker ideal L2 self ($TE = DE = -.08$; supporting hypothesis H1) and stronger ought-to L2 self ($TE = DE = .18$; supporting hypothesis H2). As L2 shame’s IEs from weaker ideal L2 self ($-.08 * .50 = -.04$) and stronger ought-to L2 self ($.18 * .22 = .0396$) mostly cancel each other ($-.04 + .0396 = -.0004$), L2 shame has no substantial total effect on IDLE. By contrast, students with more L2 guilt showed more IDLE ($TE = .50$; $DE = .19$, supporting H6), partially mediated by stronger ideal L2 self ($IE = .43 * .50 = .22$)

Table 3. Convergent and discriminant validity of L2 shame, L2 guilt, L2 motivation, and IDLE.

		AVE (> .5)		CR (> .7)		The Square root of AVE and Squared correlation coefficient matrix				
						1	2	3	4	5
1	L2 Shame	.64	.96			.80				
2	L2 Guilt	.75	.97			.10	.86			
3	Ideal L2 Self	.72	.94			-.04	.41	.85		
4	Ought L2 Self	.59	.90			.22	.39	.47	.77	
5	IDLE	.61	.96			.03	.45	.62	.47	.78

Note: The bolded values along the diagonal represent the square root of AVE.

Table 4. Model fit indices of L2 shame, L2 guilt, L2 motivation, and IDLE.

	χ^2/df	CFI	IFI	TLI	NFI	PGFI	RMSEA	SRMR
Measurement model	7.82	.92	.92	.92	.91	.75	.06	.05
Structural model	8.07	.92	.92	.91	.91	.75	.06	.07
Benchmark values (Kline 2023)	< 5	> .9	> .9	> .9	> .9	> .5	< .10	< .08

and stronger ought-to L2 self ($IE = .40 * .22 = .09$). Students with greater ideal L2 selves report more IDLE ($TE = DE = .50$), supporting H7. Likewise, students with stronger ought-to L2 selves report more IDLE ($TE = DE = .22$), supporting H8. This model yielded substantial squared multiple correlations for ideal L2 self (.19), ought-to L2 self (.19), and IDLE (.49) (Plonsky and Oswald 2014).

L2 motivation mediated these L2 emotion links with IDLE. Ideal L2 self and ought-to L2 self fully mediated the link between L2 shame and IDLE (ideal L2 self: indirect effect = .32 [lower bound = .26, upper bound = .39], $p < .001$; ought-to L2 self: indirect effect = .34 [lower bound = .28, upper bound = .41], $p < .001$). They also partially mediated the link between L2 guilt and IDLE (ideal L2 self: indirect effect = .20 [lower bound = .16, upper bound = .23], $p < .001$; ought-to L2 self: indirect effect = .08 [lower bound = .06, upper bound = .10], $p < .001$).

4.4. The effects of gender difference

Compared to men, women reported more L2 shame ($TE = DE = .07$, supporting H9a) and more L2 guilt ($TE = DE = .17$, supporting H10a) (see Figure 2). Gender moderated three of the above links (see Table 6). The negative link between L2 shame and ideal L2 self was stronger for females than for males ($\beta_{(female)} = |-0.13| > \beta_{(male)} = |-0.002|$; $p = .001$). By

Table 5. Hypotheses testing results of direct and indirect paths.

Hypotheses	β	p	t-value (>1.96)	r	Results
H1: L2 Shame → Ideal L2 Self	-.08***	$p < .001$	-3.99	-.04	Supported
H2: L2 Shame → Ought-to L2 Self	.18***	$p < .001$	8.50	.22	Supported
H3: L2 Guilt → Ideal L2 Self	.43***	$p < .001$	18.56	.42	Supported
H4: L2 Guilt → Ought-to L2 Self	.40***	$p < .001$	17.62	.41	Supported
H5: L2 Shame → IDLE	-.02 ^{n.s.}	$p = .291$	-1.06	.03	Unsupported
H6: L2 Guilt → IDLE	.19***	$p < .001$	8.93	.45	Supported
H7: Ideal L2 Self → IDLE	.50***	$p < .001$	17.98	.65	Supported
H8: Ought-to L2 Self → IDLE	.22***	$p < .001$	9.281	.50	Supported
Mediation paths	95% Confidence Interval		p (Two-tailed significance)	Indirect effect	Results
	Lower bound	Upper bound			
L2 Shame → Ideal L2 Self → IDLE	.26	.39	$p < .001$	.32	Full mediation
L2 Shame → Ought-to L2 Self → IDLE	.28	.41	$p < .001$	.34	Full mediation
L2 Guilt → Ideal L2 Self → IDLE	.16	.23	$p < .001$	.20	Partial mediation
L2 Guilt → Ought-to L2 Self → IDLE	.06	.10	$p < .001$	.08	Partial mediation

Note: $p < .001$ ***; n.s. = non-significance; $r < .25$ = small effect size, $.25 < r < .5$ = medium effect size, while $r > .5$ = large effect size (Plonsky and Ghanbar 2018; Plonsky and Oswald 2014).

Table 6. Gender moderation.

Paths	Males		Females		Group differences <i>p</i>
	Std. Regression Weights	S.E.	Std. Regression Weights	S.E.	
L2 Shame → Ideal L2 Self	−.002	.031	−.13***	.025	.001
L2 Shame → Ought-to L2 Self	.17***	.034	.17***	.026	.982
L2 Guilt → Ideal L2 Self	.50***	.042	.45***	.034	.337
L2 Guilt → Ought-to L2 Self	.60***	.044	.37***	.034	< .001
L2 Shame → IDLE	−.01	.022	−.03	.018	.541
L2 Guilt → IDLE	.21***	.033	.17***	.025	.441
Ideal L2 Self → IDLE	.47***	.039	.37***	.028	.046
Ought-to L2 Self → IDLE	.14***	.030	.17***	.023	.401

Note. p *** < .001; Model fit across groups ($\chi^2/df = 4.81$, IFI = .91, TLI = .91, CFI = .91, PGFI = .73, RMSEA = .04)

contrast, males showed stronger links between (a) L2 guilt and ought-to L2 self ($\beta_{(male)} = .60 > \beta_{(female)} = .37$; $p < .001$), and (b) ideal L2 self and IDLE ($\beta_{(male)} = .47 > \beta_{(female)} = .37$; $p = .046$).

Hence, females with greater L2 shame show less IDLE ($TE = IE = -.02$), fully mediated by weaker ideal L2 self ($IE = -.13 * .37 = -.05$) and stronger ought-to L2 self ($.17 * .17 = .03$). By contrast, males with greater L2 shame show more IDLE ($TE = IE = .02$), fully mediated by stronger ought-to L2 self ($IE = .17 * .14 = .02$).

Females with greater L2 guilt show more IDLE ($TE = .40$, $DE = .17$), partially mediated by stronger ideal L2 self ($IE = .45 * .37 = .17$) and stronger ought-to L2 self ($IE = .37 * .17 = .06$). Males with greater L2 guilt showed even more IDLE ($TE = .53$, $DE = .21$), which was partially mediated by stronger ideal L2 self ($IE = .50 * .47 = .24$) and stronger ought-to L2 self ($IE = .60 * .14 = .08$). Other links showed no significant gender differences.

5. Discussion

This study examined whether L2 emotions (L2 shame, L2 guilt), gender, and L2 motivation (ideal and ought-to L2 selves) are linked to IDLE participation among Chinese EFL undergraduates. The results showed that females reported more L2 shame and more L2 guilt than males did. Also, students with more L2 shame had weaker ideal L2 selves but stronger ought-to L2 selves. Those with more L2 guilt had greater ideal L2 selves, ought-to L2 selves, and IDLE engagement. Also, students with stronger ideal or ought-to L2 selves had more IDLE engagement. Furthermore, gender moderated the links between (a) L2 shame and ideal L2 self, (b) L2 guilt and ought-to L2 self, and (c) ideal L2 self and IDLE engagement.

5.1. RQ1: To what extent, if any, do L2 shame and L2 guilt predict L2 motivation?

L2 shame was negatively linked to the ideal L2 self. This finding supports hypothesis H1 and aligns with studies showing how L2 shame can hamper students' L2 motivation in the Iran and China (Teimouri 2018, 2019; Yan, Sun, and Shi 2024). This result also fits the view that L2 shame can exacerbate fear of errors/failure and set back their perceived L2 learning potential, and reduce their ideal L2 selves (Lewis 1971). Meanwhile, L2 shame was positively linked to ought-to L2 self among these Chinese students, echoing the positive link in Teimouri's (2018) study of Iranian students. Like Middle East Iranian students,

collectivistic Chinese students might accept and internalize L2 shame as an extrinsic motivator that reinforces external expectations from family, teachers, or society, which boosts their ought-to selves (Liu, Darvin, and Ma 2024a, 2024b). This view aligns with the importance of public self-image (*face*) in collectivist cultures (Ho and Crookall 1995), as fear of losing face from poor English performance can heighten L2 shame and transform it into an external motivational force for learning L2 (Liu et al. 2020). This result echoes Teimouri's (2019) observation among Iranian EFL learners, where those with a more salient ought-to L2 self were more susceptible to experiencing shame.

L2 guilt was positively linked to the ideal L2 self and ought-to L2 self, supporting H4 and H5. These results cohere with Teimouri's (2018) results. They also cohere with the view that EFL students who perceive more personal failure feel more guilt, accompanied by unease and remorse, which fuels their greater effort in L2 learning to align with both their internalized values/aspirations (ideal L2 self) and external social pressures to succeed (ought-to L2 self; Tangney and Dearing 2003).

5.2. RQ2: To what extent, if any, do L2 shame and L2 guilt interact with L2 motivation to predict IDLE engagement?

Among these students, those with more L2 guilt reported more IDLE, supporting H6, which was consistent with the results of prior studies on the impact of self-conscious emotions on language learning behaviors and academic involvement (Shao et al. 2023; Teimouri 2017, 2018). Specifically, Teimouri (2018) surveyed 174 Iranian tertiary students to probe the relationships between L2 shame, L2 guilt, and learning behaviors. The study's results have displayed that L2 guilt had a significant and positive impact on their communicative behaviors. To some extent, students' communicative behaviors are regarded as a way of engaging with IDLE activities. Therefore, L2 guilty students are more likely to accept responsibility for their actions, which in turn facilitates their engagement with IDLE practices. The medium effect size ($r = .45$; Plonsky and Ghanbar 2018) also supports the point that learners with higher levels of L2 guilt are more likely to engage with IDLE practices beyond the classroom. The finding also fits the view that students with a stronger sense of L2 guilt often focus attention on changeable processes and yield greater arousal (Pekrun 2006), thereby pushing them to recognize their shortcomings and try to overcome them (Teimouri 2018), thereby engaging with L2 learning activities like IDLE (Soyoof et al., 2023). By contrast, students' L2 shame was not related to IDLE participation, diverging from Teimouri's (2018) result that Iranian students feeling more L2 shame reported fewer communication behaviors. As noted above, compared to collectivistic Chinese students, Iranian students might be less accepting of L2 shame, so it might have a stronger negative effect on their self-image and communications (Teimouri 2018). Notably, the very small effect size ($r = .03$; Plonsky and Ghanbar 2018) indicates that this association should be explained cautiously. Future comparative studies can test this insight more rigorously.

Furthermore, L2 motivation not only significantly promotes IDLE engagement but also plays mediating roles in the relationships between self-conscious emotions (i.e. L2 shame and L2 guilt) and IDLE engagement. Chinese university students' ideal L2 self had a strong and positive effect on IDLE, which resonates with the findings of previous studies (Liu, Darvin, and Ma 2024a, 2024b, 2024c; Zou et al. 2025c). Additionally, the ought-to L2

self (encompassing parental and societal expectations and failure avoidance) significantly influenced IDLE engagement, which diverges from the previous studies that indicate the ought-to L2 self lacks a directly predictive power over IDLE due to potential mediating roles (e.g. Lee and Lu 2023; Liu, Darwin, and Ma 2024b). One possible explanation for this novel finding may depend on heightened social expectations and failure avoidance pressures in the Chinese context (Li, Hiver, and Papi 2022), where expectations and pressures drive students to actively seek language learning opportunities through digital devices or platforms (Liu, Zhang, and Zhang 2024c; Zou et al. 2025c).

Finally, two full mediational paths (L2 Shame $\rightarrow$ Ideal L2 Self $\rightarrow$ IDLE and L2 Shame $\rightarrow$ Ought-to L2 Self $\rightarrow$ IDLE, see Table 4) underscore that while L2 shame does not directly predict IDLE, indirect effects exist through the full mediators of ideal L2 self and ought-to L2 self. These insights not only reveal how specific motivational factors (i.e. ideal L2 self and ought-to L2 self) serve as pathways through which Chinese university students' L2 shame could be translated to their IDLE behaviors but also provide evidence that L2 learning behaviors are the outcomes of the intricate interaction of individual differences, including self-conscious emotions and motivation (Leary 2007; MacIntyre & Vincze, 2017; Teimouri 2018, 2019). Moreover, apart from fully mediating the link between L2 shame and IDLE, L2 motivation (i.e. ideal L2 self and ought-to L2 self) played partial mediating roles in the relationship between L2 guilt and IDLE engagement. This finding suggests that L2 English learners with a strong feeling of L2 guilt are not only more inclined to engage with IDLE activities but also able to develop strong senses of future selves. Therefore, we underscore that while previous research has disclosed how L2 learners with overwhelming guilt for self-promotion are more likely to conduct IDLE practices (Lewis 1971; Tangney and Dearing 2003; Teimouri 2017).

5.3. RQ3: Does gender play a moderating role in the relationships between these variables?

Gender was linked to self-conscious L2 emotions and moderated their links with other variables. The findings revealed that compared to males, females reported more L2 shame and L2 guilt. Both results are consistent with previous studies (Else-Quest et al. 2012; Kushnir et al. 2016; Lutwak and Ferrari 1996). These findings align with the view that women are more concerned about others' expectations, so when they fail to meet others' expectations, they feel more L2 shame and guilt, compared to men (Kushnir et al. 2016; Teimouri 2018, 2019). The findings also suggested that gender differences significantly moderated the relationships between certain emotional, motivational variables, and IDLE engagement. Specifically, the negative effect of L2 shame on the ideal L2 self was only significant among females ($\beta_{(female)} = -.13, p = .001$), indicating that female students experiencing a strong sense of L2 shame are more likely to reduce their idealized motivation, which aligns with the findings of Else-Quest et al. (2012) and Kushnir et al. (2016) concerning women being more sensitive to personal expectations than men. It also echoes Ferguson and Crowley's (1997) observation that, compared with general guilt-proneness, general shame proneness as a principal self-conscious emotion emerges more strongly among females than males. Second, compared to females, L2 trait-like guilt proneness had a stronger positive effect on the ought-to L2 self among males ($\beta_{(male)} = .60$ vs. $\beta_{(female)} = .37, p < .001$), being in line with the observation by

Kushnir et al. (2016) in terms of autonomous motivation linked with higher trait-like guilt proneness for males. It is consistent with Eisenchlas's (2013) insights that males with a stronger sense of guilt proneness are more likely than females to take actions in order to meet their parental or social expectations. Third, the SEM analysis reveal that while both male and female students benefited from a strong ideal L2 self in predicting their IDLE engagement, the effect was significantly stronger among males ($\beta_{\text{(male)}} = .47$ vs. $\beta_{\text{(female)}} = .37$, $p = .046$). Though it demonstrated the small effect discrepancy between genders relating to IDLE engagement, its implications are far from trivial. For instance, given that IDLE engagement encompasses self-directed vocabulary learning activities and incidental exposure to new vocabularies (e.g. watching English movies or videos, chatting in English with foreign friends in social media platforms, and understanding how to play games by learning new vocabularies), even a small gender difference in the motivational catalyst could accumulate over time to make meaningful impacts in L2 learning performance (Lai, Zhu, and Gong 2015; Lai and Wang 2025; Sundqvist and Nilsson 2024). Such repeated exposure to new words helps them learn, use and consolidate vocabulary knowledge for authentic communication. Concerning other relationships, gender differences did not play moderating roles.

5.4. Pedagogical implications

Based on these insights, this study yields several pedagogical implications. First, given the direct and negative impact of L2 shame on the ideal L2 self, it behoves educators and language teachers to create supportive and inclusive learning environments and embolden students to embrace challenges without apprehension of judgment or derision. For example, teachers can implement a 'mistake of week' activity, where students share the language mistakes they made and what they learned from them, thereby normalizing the language learning process and fostering a vivid ideal self-image and aspirations (Teimouri 2019). In addition, recognizing the power of L2 shame as a strong motivator for triggering ought-to L2 self, teachers may consider encouraging positive peer influence by designing collaborative assignments or undertakings in which students cooperate to achieve a shared linguistic goal. To exemplify, instructors may allocate group English presentations necessitating individual learner participation, thereby nurturing a sense of obligation to engage in assiduous preparation to circumvent disappointing their peers (Teimouri 2017). Second, concerning the positive and direct impact of L2 guilt on the ideal L2 self and the ought-to L2 self, educators may incorporate regular self-reflection and goal-setting activities that prompt students to examine their language learning progress, goals, and aspirations (Teimouri 2019). For example, teachers can invite students to compose prospective self-reflections, articulating their envisioned English-speaking self-images and delineating the steps they will undertake to actualize these aspirations. Third, recognizing the indirect and positive effect of L2 shame on IDLE practices through the ideal L2 self and the ought-to L2 self, teachers may reframe students' L2 shame as a catalyst for elucidating their English-related self-perceptions and goals, empowering them to transmute this discomfort into behaviors that mirror their desired L2 self-images, particularly within informal digital learning contexts. Finally, given the indirect and positive impact of L2 guilt on IDLE activities through the ideal and ought-to L2 selves, educators may capitalize on guilt and reframe students' L2 guilt as a

chance for redemption. For example, teachers can say something like, 'It's never too late to improve. What action can you take today to move a step closer to your future, fluent English-speaking self?' Such encouragement can help students appreciate the importance of their actions and motivate them to engage with IDLE (Dressman and Sadler 2020).

6. Conclusion, limitations, and future research

This study contributes to the understanding of how Chinese EFL undergraduate students' under-researched self-conscious emotions (i.e. L2 shame and L2 guilt) negotiate with their L2 motivation (i.e. ideal L2 self and ought-to L2 self) to influence their IDLE engagement. The research of undergraduates learning English showed that compared to men, women reported a stronger sense of L2 shame and L2 guilt. Among these students, those with greater L2 shame had stronger ought-to L2 selves but weaker ideal L2 selves. Meanwhile, those with more L2 guilt had stronger ought-to and ideal L2 selves, yielding more IDLE engagement. Compared to males, females' L2 guilt was less positively linked to their ought-to L2 selves, and their L2 shame was more negatively linked to their ideal L2 selves and IDLE practices. This study also extends a pattern of understanding how gender differences play moderating roles among the interrelationships between L2 shame, L2 guilt, ideal L2 self, ought-to L2 self, and IDLE engagement.

Moreover, we believe that the contributions of the study are twofold. First, it investigates how Chinese EFL learners' self-conscious emotions and motivational factors influence their IDLE engagement. In contrast to previous studies that primarily explored the relationships between self-conscious emotions and language achievement, our study offers unique insights in understanding how L2 learners' educational psychological factors affect their digital language learning behaviors beyond the classroom. Second, it uncovers gender differences across emotional, motivational, and technological factors. Compared with prior literature, our work did the first attempt to reveal how gender difference affects self-conscious emotions (L2 shame and L2 guilt) and IDLE engagement and moderate the relationships between emotional, motivational, and technological variables in Second Language Acquisition (SLA) field. It extends theoretical insights into the complex associations of emotional, motivational aspects of digital language learning from a gender difference perspective.

Despite the fact that these contributions are significant, this study has five limitations that future research should address: a homogenous sample, cross-sectional data, only survey data, inadequate contextual data, and without examination of language proficiency levels. First, this sample only included undergraduates from one country, so future studies can include diverse participants across ages, regions, and cultures. Second, this cross-sectional data cannot capture the dynamic ebb and flow of IDLE participation over time. So, we need longitudinal studies to grasp how students navigate autonomous language learning with the affordance of technology. Third, as this study only included survey data, future research can include other data like videotapes, interviews, or simulations in a mixed-methods design to uncover the intricate patterns of students' emotions and motivations during IDLE participation. Fourth, students' engagement with IDLE might differ across contexts. Learning atmosphere, peer expectations, or teacher support might affect students' IDLE engagement. Thus, future research can consider such elements to offer a more comprehensive view of how L2 students negotiate

their contexts, emotions, and motivations to engage with IDLE activities. Lastly, the present study did not measure participants' language proficiency levels. Given that variation in learners' proficiency levels may influence their emotional experiences in language learning, we call for future research that should incorporate systematic measures of language proficiency to better capture its potential effect.

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Authors' contribution

All authors have participated in research and article preparation. In addition, all authors have approved the final article.

Consent to participate

Informed consent to participate was obtained from all individual participants included in the study.

Data availability statement

The raw data can be made available upon request.

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Appendix

A. L2 Shame scale (Teimouri, 2018)

(13-item scale, Likert 6-point) (1 Strongly disagree – 6 Strongly agree)

L2 shame

Definition: L2 shame is defined as a self-conscious, negative emotional experience that arises when L2 learners perceive a discrepancy between their actual and ideal L2 self. During the L2 shame experience, learners may critically direct their attention toward themselves, negatively scrutinize every aspect of it, and harshly perceive themselves as incompetent, flawed language users.

- S1 During a class discussion, you can't express your ideas in English, so you express them in Chinese. The teacher reminds you that you need to speak in English. I would feel incompetent.
 - S2 You think your writing assignment is good. The teacher corrects your writing and returns it to you. You realize you have made many mistakes. I would feel incompetent writing in English.
 - S3 At the end of the term, you realize that your progress in learning English is not satisfactory. I would feel hopeless and helpless.
 - S4 In class, the teacher divides students into several groups. She hands out a copy of your writing to each group and asks them to correct it. Students pick up many mistakes in your writing. I would feel weak and not confident.
 - S5 In class, the teacher asks you a question in English. You don't understand the question. The teacher asks the question in a different way. Again, you don't understand the question. I would feel dumb.
 - S6 In class, students are having a group discussion about a topic. Although you have some good ideas to share, you are worried that you can't express them properly in English. You remain quiet and avoid participating in the discussion. I would feel like I am a coward.
 - S7 In class, students are discussing a topic in English. You can't express your ideas well in English. Nobody understands what you mean. I would feel my English proficiency is worse than the other students'.
 - S8 You receive your English class report card. You are surprised to see that your grades are lower than what you expected. I would feel I don't have the talent to learn English.
 - S9 The class will be working on your writing assignment today. However, you have forgotten to write it and bring it to class. I would think, "I am an irresponsible person."
 - S10 In class, students are asked to read an English text and answer the questions within a period of time. While all the students have completed the task, you can't finish reading the text. Therefore, you can't answer some of the questions. I would fear humiliation if suddenly the teacher asked me a question.
 - S11 In class, you are speaking about something in English. The teacher interrupts you to correct your grammatical mistakes. I would think other students are looking at me and laughing.
 - S12 You are home, and an English program is playing on TV. Your family members ask you to translate some parts of the program. You can't translate them well. I would feel small.
 - S13 In class, the teacher plays an audio recording for today's lesson. After listening to it several times, you understand just a few points and can't answer most of the questions. I would feel my English proficiency is worse than the other students'.
-

B. L2 Guilt scale (Teimouri, 2018)

(13-item scale, Likert 6-point) (1 Strongly disagree – 6 Strongly agree)

L2 guilt

Definition: L2 guilt refers to a self-conscious emotion experienced by L2 learners when they perceive their language-related behaviors as falling short of personal and social expectations. During the L2 guilt experience, learners may critically direct their attention toward their behaviors and scrutinize ways to undo them.

- G1 You receive your English class report card. You are surprised to see that your grades are lower than what you expected. I should have studied more; I would try to do better in the future.
- G2 You think your writing assignment is good. The teacher corrects your writing and returns it to you. You realize you have made many mistakes. I would try to write more carefully next time.
- G3 In class, the teacher divides students into several groups. She hands out a copy of your writing to each group and asks them to correct it. Students pick up many mistakes in your writing. I should have been more careful and spent more time on my writing.
- G4 In class, you are speaking about something in English. The teacher interrupts you to correct your grammatical mistakes. I would try to avoid repeating those mistakes.
- G5 At the end of the term, you realize that your progress in learning English is not satisfactory. I would feel sad and work harder next term.
- G6 During a class discussion, you can't express your ideas in English, so you express them in Chinese. The teacher reminds you that you need to speak in English. I would apologize and try to express my ideas in English.
- G7 The class will be working on your writing assignment today. However, you have forgotten to write it and bring it to class. I would apologize and promise it wouldn't happen again.
- G8 In class, students are asked to read an English text and answer the questions within a period of time. While all the students have completed the task, you can't finish reading the text. Therefore, you can't answer some of the questions. I would try to boost my reading speed by studying more.
- G9 In class, students are discussing a topic in English. You can't express your ideas well in English. Nobody understands what you mean. I would try harder next time to state what I mean more clearly.
- G10 In class, students are having a group discussion about a topic. Although you have some good ideas to share, you are worried that you can't express them properly in English. You remain quiet and avoid participating in the discussion. I would feel regret; I would speak my mind next time.
- G11 You are home, and an English program is playing on TV. Your family members ask you to translate some parts of the program. You can't translate them well. I would feel sad and try to improve my English proficiency.
- G12 In class, the teacher asks you a question in English. You don't understand the question. The teacher asks the question in a different way. Again, you don't understand the question. I would politely ask the teacher to repeat her question.
- G13 In class, the teacher plays an audio recording for today's lesson. After listening to it several times, you understand just a few points and can't answer most of the questions. I wish I had practiced the audio before.

Note: The underlined text refers to the situations experienced by the students.

C. L2 Motivation scale (Dornyei, 2009; You et al., 2016)

(12-item scale, Likert 6-point) (1 Strongly disagree – 6 Strongly agree)

Ideal L2 self

Definition: Ideal L2 self represents the motives growing out of personal desires for growth, accomplishments, and positive outcomes associated with learning ENGLISH.

- IS1 Whenever I think about my future career, I imagine myself using English.
- IS2 I can imagine myself doing business with foreigners through English.
- IS3 I can imagine myself speaking English with international friends or colleagues.
- IS4 I can imagine myself as a person who is able to speak English.
- IS5 I can imagine myself talking in English with foreign friends in informal situations in the future.
- IS6 I can imagine myself having discussions in English when living or travelling abroad.

Ought-to L2 self

Definition: Ought-to L2 self comprises the motives related to the attempts to meet external expectations and avoid negative consequences that may arise from not learning ENGLISH.

- OS1 If I don't learn English, it will have a negative impact on my life.
- OS2 Learning English is important to me because people around me think I should learn it.
- OS3 Learning English is important to me because it will give me social recognition.
- OS4 Learning English is important to me because I want to gain recognition from my teachers and classmates.
- OS5 Learning English is important to me because an educated person should be able to use English.
- OS6 Learning English is important to me because other people will respect me more if I have knowledge of English.

D. Form-focused and Meaning-focused informal digital learning of English scale (FM-IDLES) bilingual version (Authors, 2025c)

(14-item scale, Likert 6-point) (1 Strongly disagree – 6 Strongly agree)

Form-focused informal digital learning of English activities (FIA)*Definition:* FIA refers to out-of-class and technology-mediated English learning activities that primarily target the formal aspects of language, such as grammar, vocabulary, pronunciation, and sentence structure.

- F1 I actively listen to English news or podcasts on my mobile phone or computer outside class to improve my English listening skills.
- F2 I actively use English learning apps to practise my oral expression and improve my English-speaking skills outside the classroom.
- F3 I actively use English learning apps to practice and improve my English pronunciation outside the classroom.
- F4 I actively improve my English reading skills by reading English articles/contents through online resources outside the classroom.
- F5 I actively use mobile phone apps or websites for online English writing practice outside of class to improve my English writing skills.
- F6 I actively use vocabulary apps outside of class to expand my English vocabulary.
- F7 I actively use mobile phone apps or websites to learn English grammar outside of class.

Meaning-focused informal digital learning of English activities (MIA)*Definition:* MIA involves out-of-class and technology-mediated English learning that emphasizes comprehension, communication, and the practical application of language in real-life contexts.

- M1 I actively watch short English videos (e.g. vlogs) on social media outside the classroom.
 - M2 I actively communicate with other learners in English through social media apps outside the classroom.
 - M3 I actively chat with foreigners in English through social media software outside the classroom.
 - M4 I actively read English articles online or English-related posts on social media.
 - M5 I actively participate in online English discussion platforms to discuss with others on topics that are currently popular or familiar to me.
 - M6 I actively watch English-language films and TV series outside the classroom.
 - M7 I actively search online (either in Chinese or English) to learn about the culture and history of English-speaking countries/languages to develop my intercultural communication skills.
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